

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10		FA1								812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.
20										816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.
20							*****VALVES*****				
30											
30		VAT					***** MODULAR VALVES *****			817	USE ONLY FOR TIE IN
40										B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
40		VAV, 817	MODULAR DBB	K10CL16	2	3	MODULAR VALVE DB&B BALL TYPE 10000 CF INCONEL 625 BODY AND TRIM METAL GO FULL SIZE TYPE 2BV	0	0		
50		VAV, 817	MODULAR DBB	K10PL16	2	2	MODULAR VALVE DB&B BALL TYPE 10000 CF INCONEL 625 BODY AND TRIM METAL GO FULL SIZE TYPE 2BV REDUCING FLANGES	3	3	BA3	TO BE USED WITH API 6A FLANGES ONLY.
60	U04	817	MODULAR DBB	K10CL16L	2	3	MODULAR VALVE DB&B BALL TYPE 10000 CF INCONEL 625 BODY AND TRIM METAL GO FULL SIZE TYPE 2BV	0	0	BA5	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY. FOR FGP ONLY.
70	U04	VAX, 817	MODULAR DBB	M10CL14	2	2	MODULAR VALVE DB&B BALL TYPE 10000 CF INCONEL 625 BODY AND TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	FA1	API 6BX FLANGE TO NB PIPE DESIGNATION SEE PIM-SU-5112-TCO
80	U04	817	MODULAR DBB	K10TL16	3	3	MODULAR VALVE DB&B BALL TYPE 10000 CF X RTJ (BX TYPE) INCONEL 625 BODY AND TRIM METAL GO FULL SIZE TYPE 2BV	0	0	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.
90	U04	817	MODULAR DBB	K10TL16L	3	3	MODULAR VALVE DB&B BALL TYPE 10000 CF X RTJ (BX TYPE) INCONEL 625 BODY AND TRIM METAL GO FULL SIZE TYPE 2BV	0	0	VAT	CONTROLLED BORE (2BVR) TYPE VALVE TO BE USED FOR INSTRUMENT ISOLATION ONLY
100	U04	VAX, 817	MODULAR DBB	M10TL14	2	2	MODULAR VALVE DB&B BALL TYPE 10000 CF X RTJ (BX TYPE) INCONEL 625 BODY AND TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
110	U04	VAX, 817	MODULAR DBB	M10TL14L	2	2	MODULAR VALVE DB&B BALL TYPE 10000 CF X RTJ (BX TYPE) INCONEL 625 BODY AND TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	VAV	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
120	U04	VAX, 817	MODULAR DBB	M10UL14	2	2	MODULAR VALVE DB&B BALL TYPE 10000 RTJ (BX TYPE) X CF INCONEL 625 BODY AND TRIM METAL LO - CONTROLLED BORE TYPE R2BVR	3	3	VAX	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
130	U04	VAX, 817	MODULAR DBB	M10UL14L	2	2	MODULAR VALVE DB&B BALL TYPE 10000 RTJ (BX TYPE) X CF INCONEL 625 BODY AND TRIM METAL LO - CONTROLLED BORE TYPE R2BVR	3	3	VAX	RTJ(BX TYPE) END IS SIDE MATING WITH INSTRUMENTATION
140							***** BALL VALVES *****				
140											
150											
150		VAV, 817	BALL	B10CL16	0.75	3	BALL 10000 CF INCONEL 625 BODY AND TRIM TRUNNION METAL FB GO	0	0	VC7	10000PSI RATED CHECK VALVES SHALL BE SP ITEMS. REFER TO SP LIST FOR DETAILS
160	U04	817	BALL	B10CH16	4	22	BALL 10000 CF F22 INCONEL 625 TRIM TRUNNION METAL FB GO	0	0		
170			BALL	B10LL16	0.75	3	BALL 10000 API 6A RJ INCONEL 625 BODY AND TRIM TRUNNION METAL FB GO	0	0		
180			BALL	B10LH16	4	22	BALL 10000 API 6A RJ F22 INCONEL 625 TRIM TRUNNION METAL FB GO	0	0		
190											
190							***** THROUGH CONDUIT GATE VALVES *****				
200											
200	U04	817	GATE	W10CL72	2	2	GATE 10000 CF INCONEL 625 BODY AND TRIM THROUGH CONDUIT SOLID SLAB HW	0	0		
210	U04	817	GATE	W10CH75	3	3	GATE 10000 CF F22 TRIM INC 625 THROUGH CONDUIT SOLID SLAB GO	0	0		
220	U04	817	GATE	W10CH73	4	12	GATE 10000 CF F22 TRIM INC 625 THROUGH CONDUIT SOLID SLAB GO	0	0		
230											
230							***** GLOBE VALVES *****				
240											
240	U04	817	GLOBE	L10CL43	2	2	GLOBE 10000# CF INC 625 B564 INC 625 TRIM OS&Y HW	0	0		
250											
250		VC7					***** CHECK VALVES *****				
260											
260							*****PIPE*****				
270											
270		812	PIPE	5358511	0.75	1	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0		
280			PIPE	5358541	1.5	1.5	PIPE WT 12.0MM SMLS LTCS A333 GR.6 PE	0	0		
290			PIPE	5360062	2	2	PIPE WT 10.5MM SMLS 2-1/4CR F22M BE	0	0		
300			PIPE	5360067	3	3	PIPE WT 14.0MM SMLS 2-1/4CR F22M BE	0	0		
310			PIPE	5360070	4	4	PIPE WT 17.0MM SMLS 2-1/4CR F22M BE	0	0		
320			PIPE	5360071	6	6	PIPE WT 23.5MM SMLS 2-1/4CR F22M BE	0	0		
330			PIPE	5360075	8	8	PIPE WT 29.5MM SMLS 2-1/4CR F22M BE	0	0		
340			PIPE	5360079	10	10	PIPE WT 36.0MM SMLS 2-1/4CR F22M BE	0	0		
350			PIPE	5360084	12	12	PIPE WT 42.0MM SMLS 2-1/4CR F22M BE	0	0		
360			PIPE	5360085	14	14	PIPE WT 46.0MM SMLS 2-1/4CR F22M BE	0	0		
370			PIPE	5360088	16	16	PIPE WT 52.0MM SMLS 2-1/4CR F22M BE	0	0		
380			PIPE	5360090	18	18	PIPE WT 58.0MM SMLS 2-1/4CR F22M BE	0	0		
390			PIPE	5360092	20	20	PIPE WT 64.5MM SMLS 2-1/4CR F22M BE	0	0		
400			PIPE	5360095	22	22	PIPE WT 70.5MM SMLS 2-1/4CR F22M BE	0	0		
410											
410							***** FITTINGS *****				
420											
420			ELBOW	5392870	0.75	1	ELL 90 DEG LR XXS LTCS A420 WPL6	0	0		
430			ELBOW	5392871	1.5	1.5	ELL 90 DEG LR WT 12.0MM LTCS A420 WPL6	0	0		
440			ELBOW	5406636	2	2	ELL 90 DEG LR WT 10.5MM 2-1/4CR F22M	0	0		
450			ELBOW	5406637	3	3	ELL 90 DEG LR WT 14.0MM 2-1/4CR F22M	0	0		
460			ELBOW	5406600	4	4	ELL 90 DEG LR WT 17.0MM 2-1/4CR F22M	0	0		
470			ELBOW	5406601	6	6	ELL 90 DEG LR WT 23.5MM 2-1/4CR F22M	0	0		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - NONE	CORR. ALLOW. 3 MM	SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN CODE ASME B31.3 CHAPTER IX / API 6A DESIGN PRESSURE - 600 BARG FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 10000 RJ/CF TEMPERATURE RANGE:- 40 / 130 °C	12/16/2024	BY OZAC	U04	Page 1 of 5	10000F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES
480			ELBOW	5406605	8	8	ELL 90 DEG LR WT 29.5MM 2-1/4CR F22M	0 0	
490			ELBOW	5406609	10	10	ELL 90 DEG LR WT 36.0MM 2-1/4CR F22M	0 0	
500			ELBOW	5406614	12	12	ELL 90 DEG LR WT 42.0MM 2-1/4CR F22M	0 0	
510			ELBOW	5406615	14	14	ELL 90 DEG LR WT 46.0MM 2-1/4CR F22M	0 0	
520			ELBOW	5406618	16	16	ELL 90 DEG LR WT 52.0MM 2-1/4CR F22M	0 0	
530			ELBOW	5406620	18	18	ELL 90 DEG LR WT 58.0MM 2-1/4CR F22M	0 0	
540			ELBOW	5406622	20	20	ELL 90 DEG LR WT 64.5MM 2-1/4CR F22M	0 0	
550			ELBOW	5406635	22	22	ELL 90 DEG LR WT 70.5MM 2-1/4CR F22M	0 0	
560			ELBOW	5392876	0.75	1	ELL 45 DEG LR XXS LTCS A420 WPL6	0 0	
570			ELBOW	5392875	1.5	1.5	ELL 45 DEG LR WT 12.0MM LTCS A420 WPL6	0 0	
580			ELBOW	5407083	2	2	ELL 45 DEG LR WT 10.5MM 2-1/4CR F22M	0 0	
590			ELBOW	5407084	3	3	ELL 45 DEG LR WT 14.0MM 2-1/4CR F22M	0 0	
600			ELBOW	5407060	4	4	ELL 45 DEG LR WT 17.0MM 2-1/4CR F22M	0 0	
610			ELBOW	5407061	6	6	ELL 45 DEG LR WT 23.5MM 2-1/4CR F22M	0 0	
620			ELBOW	5407065	8	8	ELL 45 DEG LR WT 29.5MM 2-1/4CR F22M	0 0	
630			ELBOW	5407069	10	10	ELL 45 DEG LR WT 36.0MM 2-1/4CR F22M	0 0	
640			ELBOW	5407074	12	12	ELL 45 DEG LR WT 42.0MM 2-1/4CR F22M	0 0	
650			ELBOW	5407075	14	14	ELL 45 DEG LR WT 46.0MM 2-1/4CR F22M	0 0	
660			ELBOW	5407078	16	16	ELL 45 DEG LR WT 52.0MM 2-1/4CR F22M	0 0	
670			ELBOW	5407080	18	18	ELL 45 DEG LR WT 58.0MM 2-1/4CR F22M	0 0	
680			ELBOW	5407082	20	20	ELL 45 DEG LR WT 64.5MM 2-1/4CR F22M	0 0	
690			ELBOW	5407095	22	22	ELL 45 DEG LR WT 70.5MM 2-1/4CR F22M	0 0	
700			ELBOW	5407045	8	8	ELL 22.5 DEG LR WT 29.5 MM 2-1/4CR F22M	0 0	
710									
710			TEE	5392879	0.75	1	TEE SCH XXS LTCS A420 WPL6	0 0	
720			TEE	5392889	1.5	1.5	TEE WT 12.0MM LTCS A420 WPL6	0 0	
730			TEE	5407180	2	2	TEE WT 10.5MM 2-1/4CR F22M	0 0	
740			TEE	5407181	3	3	TEE WT 14.0MM 2-1/4CR F22M	0 0	
750			TEE	5407160	4	4	TEE WT 17.0MM 2-1/4CR F22M	0 0	
760			TEE	5407161	6	6	TEE WT 23.5MM 2-1/4CR F22M	0 0	
770			TEE	5407165	8	8	TEE WT 29.5MM 2-1/4CR F22M	0 0	
780			TEE	5407169	10	10	TEE WT 36.0MM 2-1/4CR F22M	0 0	
790			TEE	5407173	12	12	TEE WT 42.0MM 2-1/4CR F22M	0 0	
800			TEE	5407174	14	14	TEE WT 46.0MM 2-1/4CR F22M	0 0	
810			TEE	5407177	16	16	TEE WT 52.0MM 2-1/4CR F22M	0 0	
820			TEE	5407178	18	18	TEE WT 58.0MM 2-1/4CR F22M	0 0	
830			TEE	5407213	22	22	TEE WT 70.5MM 2-1/4CR F22M	0 0	
840	U04		TEE RED	5392877	0.75	0.75	TEE RED SCH XXS LTCS A420 WPL6	1 1	
850	U04		TEE RED	5392887	0.75	1	TEE RED WT 12.0MM / SCH XXS LTCS A420 WPL6	1.5 1.5	
860	U04		TEE RED	5407366	0.75	1	TEE RED SCH 10.5MM / XXS 2-1/4CR F22M	2 2	
870	U04		TEE RED	5407367	1.5	1.5	TEE RED BW WT 10.5MM / 12.0MM 2-1/4CR F22M	2 2	
880	U04		TEE RED	5407368	1.5	1.5	TEE RED BW 14.0MM / 12.0MM 2-1/4CR F22M	3 3	
890	U04		TEE RED	5407369	2	2	TEE RED BW WT 14.0MM / 10.5MM 2-1/4CR F22M	3 3	
900	U04		TEE RED	5407370	2	2	TEE RED BW SCH 17.0MM / 10.5MM 2-1/4CR F22M	4 4	
910	U04		TEE RED	5407371	3	3	TEE RED BW WT 17.0MM / 14.0MM 2-1/4CR F22M	4 4	
920	U04		TEE RED	5407372	3	3	TEE RED BW WT 23.5MM / 14.0MM 2-1/4CR F22M	6 6	
930	U04		TEE RED	5407381	6	6	TEE RED BW WT 29.5MM / 23.5MM 2-1/4CR F22M	8 8	
940	U04		TEE RED	5407403	6	6	TEE RED BW WT 46.0MM / 23.5MM 2-1/4CR F22M	14 14	
950	U04		TEE RED	5407365	8	8	TEE RED BW WT 46.0MM / 29.5MM 2-1/4CR F22M	14 14	
960	U04		TEE RED	5407395	10	10	TEE RED BW WT 46.0MM / 36.0MM 2-1/4CR F22M	14 14	
970	U04		TEE RED	5407373	6	6	TEE RED BW WT 52.0MM / 23.5MM 2-1/4CR F22M	16 16	
980	U04		TEE RED	5407376	8	8	TEE RED BW WT 52.0MM / 29.5MM 2-1/4CR F22M	16 16	
990	U04		TEE RED	5407392	10	10	TEE RED BW WT 52.0MM / 36.0MM 2-1/4CR F22M	16 16	
1000	U04		TEE RED	5407374	14	14	TEE RED BW WT 52.0MM / 46.0MM 2-1/4CR F22M	16 16	
1010	U04		TEE RED	5407375	8	8	TEE RED BW WT 58.0MM / 29.5MM 2-1/4CR F22M	18 18	
1020	U04		TEE RED	5407378	14	14	TEE RED BW WT 58.0MM / 46.0MM 2-1/4CR F22M	18 18	
1030	U04		TEE RED	5407377	16	16	TEE RED BW WT 58.0MM / 52.0MM 2-1/4CR F22M	18 18	
1040	U04		TEE RED	5407379	18	18	TEE RED BW WT 70.5MM / 58.0MM 2-1/4CR F22M	22 22	
1050									
1050	U04		CAP	5392880	0.75	1	CAP BW SCH XXS LTCS A420 WPL6	0 0	
1060			CAP	5392883	1.5	1.5	CAP WT 12.0MM LTCS A420 WPL6	0 0	
1070			CAP	5407124	2	2	CAP WT 10.5MM 2-1/4CR F22M	0 0	
1080			CAP	5407125	3	3	CAP WT 14.0MM 2-1/4CR F22M	0 0	
1090			CAP	5407100	4	4	CAP WT 17.0MM SMLS 2-1/4CR F22M	0 0	
1100			CAP	5407101	6	6	CAP WT 23.5MM SMLS 2-1/4CR F22M	0 0	
1110			CAP	5407105	8	8	CAP WT 29.5MM SMLS 2-1/4CR F22M	0 0	
1120			CAP	5407109	10	10	CAP WT 36.0MM SMLS 2-1/4CR F22M	0 0	
1130			CAP	5407114	12	12	CAP WT 42.0MM SMLS 2-1/4CR F22M	0 0	
1140			CAP	5407115	14	14	CAP WT 46.0MM SMLS 2-1/4CR F22M	0 0	
1150			CAP	5407118	16	16	CAP WT 52MM SMLS 2-1/4CR F22M	0 0	
1160			CAP	5407120	18	18	CAP WT 58.0MM SMLS 2-1/4CR F22M	0 0	
1170			CAP	5407122	20	20	CAP WT 64.5MM SMLS 2-1/4CR F22M	0 0	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - NONE	CORR. ALLOW. 3 MM	SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN CODE ASME B31.3 CHAPTER IX / API 6A DESIGN PRESSURE - 600 BARG FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 10000 RJ/CF TEMPERATURE RANGE: -40 / 130 °C	12/16/2024	BY OZAC	U04	Page 2 of 5	10000F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1180			CAP	5407153	22	22	CAP WT 70.5MM 2-1/4CR F22M	0	0	
1190										
1190			RED CONC	5392881	0.75	0.75	REDUCER CONC SCH XXS LTCS A420 WPL6	1	1	
1200			RED CONC	5392884	0.75	1	REDUCER CONC WT 12.0MM / SCH XXS LTCS A420 WPL6	1.5	1.5	
1210			RED CONC	5407231	0.75	1	REDUCER CONC WT 10.5MM/SCH XXS 2-1/4CR F22M	2	2	
1220			RED CONC	5407232	1.5	1.5	REDUCER CONC WT 10.5MM/12.0MM 2-1/4CR F22M	2	2	
1230			RED CONC	5407233	1.5	1.5	REDUCER CONC WT 14.0MM/12.0MM 2-1/4CR F22M	3	3	
1240			RED CONC	5407234	2	2	REDUCER CONC WT 14.0MM/10.5MM 2-1/4CR F22M	3	3	
1250			RED CONC	5407236	2	2	REDUCER CONC WT 17.0MM/10.5MM 2-1/4CR F22M	4	4	
1260			RED CONC	5407235	4	4	REDUCER CONC WT 29.5MM/17.0MM 2-1/4CR F22M	8	8	
1270			RED CONC	5407255	6	6	REDUCER CONC WT 29.5MM/23.5MM 2-1/4CR F22M	8	8	
1280			RED CONC	5407278	6	6	REDUCER CONC WT 36.0MM/ 23.5MM 2-1/4CR F22M	10	10	
1290			RED CONC	5407253	8	8	REDUCER CONC WT 46.0MM/29.5MM 2-1/4CR F22M	14	14	
1300			RED CONC	5407256	8	8	REDUCER CONC WT 52.0MM/29.5MM 2-1/4CR F22M	16	16	
1310			RED CONC	5407254	14	14	REDUCER CONC WT 52.0MM/46.0MM 2-1/4CR F22M	16	16	
1320			RED CONC	5407258	14	14	REDUCER CONC WT 58.0MM/46.0MM 2-1/4CR F22M	18	18	
1330			RED CONC	5407298	12	12	REDUCER CONC WT 52.0MM/42.0MM 2-1/4CR F22M	16	16	
1340			RED CONC	5407257	16	16	REDUCER CONC WT 58.0MM/52.0MM 2-1/4CR F22M	18	18	
1350			RED CONC	5407259	18	18	REDUCER CONC WT 70.5MM/58.0MM 2-1/4CR F22M	22	22	
1360			RED ECC	5392882	0.75	0.75	REDUCER ECC SCH XXS LTCS A420 WPL6	1	1	
1370			RED ECC	5392885	0.75	1	REDUCER ECC WT 12.0MM / SCH XXS LTCS A420 WPL6	1.5	1.5	
1380			RED ECC	5407345	0.75	1	REDUCER ECC 10.5MM / SCH XXS 2-1/4CR F22M	2	2	
1390			RED ECC	5407346	1.5	1.5	REDUCER ECC 10.5MM / 12.0MM 2-1/4CR F22M	2	2	
1400			RED ECC	5407347	1.5	1.5	REDUCER ECC 14.0MM / 12.0MM 2-1/4CR F22M	3	3	
1410			RED ECC	5407348	2	2	REDUCER ECC 14.0MM / 10.5MM 2-1/4CR F22M	3	3	
1420			RED ECC	5407349	2	2	REDUCER ECC 17.0MM / 10.5MM 2-1/4CR F22M	4	4	
1430			RED ECC	5407312	6	6	REDUCER ECC WT 29.5MM/23.5MM 2-1/4CR F22M	8	8	
1440	U04		RED ECC	5407352	4	4	REDUCER ECC WT 29.5MM/17.0MM 2-1/4CR F22M	8	8	
1450			RED ECC	5407313	6	6	REDUCER ECC WT 36.0MM/23.5MM 2-1/4CR F22M	10	10	
1460			RED ECC	5407338	8	8	REDUCER ECC WT 36.0MM/ 29.5MM 2-1/4CR F22M	10	10	
1470			RED ECC	5407321	8	8	REDUCER ECC WT 42.0MM/ 29.5MM 2-1/4CR F22M	12	12	
1480			RED ECC	5407311	8	8	REDUCER ECC WT 46.0MM/29.5MM 2-1/4CR F22M	14	14	
1490			RED ECC	5407315	8	8	REDUCER ECC WT 52.0MM/29.5MM 2-1/4CR F22M	16	16	
1500			RED ECC	5407310	10	10	REDUCER ECC WT 52.0MM/36.0MM 2-1/4CR F22M	16	16	
1510			RED ECC	5407314	14	14	REDUCER ECC WT 52.0MM/46.0MM 2-1/4CR F22M	16	16	
1520			RED ECC	5407318	14	14	REDUCER ECC WT 58.0MM/46.0MM 2-1/4CR F22M	18	18	
1530			RED ECC	5407316	12	12	REDUCER ECC WT 58.0MM/42.0MM 2-1/4CR F22M	18	18	
1540			RED ECC	5407317	16	16	REDUCER ECC WT 58.0MM/52.0MM 2-1/4CR F22M	18	18	
1550			RED ECC	5407319	18	18	REDUCER ECC WT 70.5MM/58.0MM 2-1/4CR F22M	22	22	
1560							***** EXTRUDED BRANCHES *****			
1570		816					***** FLANGES *****			
1580										
1580	U04	817	FLANGE	5851252	2	2	FLANGE COMPACT 10000# RJ WITH 9/16" FEMALE AUTOCLAVE CONNECTION 2-1/4CR F22M	0	0	
1590										
1590	U04	817	FLANGE	5619724	0.5	0.5	FLG COMPACT 10000# RJ BORE 6.4MM LTCS A350 LF2 CL.1 LINE CLASS 10000F22	0	0	
1600	U04	817	FLANGE	5619871	0.75	0.75	FLG COMPACT 10000# RJ BORE 11.0MM LTCS A350 LF2 CL.1	0	0	
1610	U04	817	FLANGE	5619872	1	1	FLG COMPACT 10000# RJ BORE 15.2MM LTCS A350 LF2 CL.1	0	0	
1620	U04	817	FLANGE	5619873	1.5	1.5	FLG COMPACT 10000# RJ BORE 24.3MM LTCS A350 LF2 CL.1	0	0	
1630	U04	817	FLANGE	5619874	2	2	FLG COMPACT 10000# RJ BORE 39.3MM 2-1/4CR F22M	0	0	
1640	U04	817	FLANGE	5619875	3	3	FLG COMPACT 10000# RJ BORE 60.9MM 2-1/4CR F22M	0	0	
1650	U04	817	FLANGE	5619896	4	4	FLG COMPACT 10000# RJ BORE 80.3MM 2-1/4CR F22M	0	0	
1660	U04	817	FLANGE	5619897	6	6	FLG COMPACT 10000# RJ BORE 121.3MM 2-1/4CR F22M	0	0	
1670	U04	817	FLANGE	5619876	8	8	FLG COMPACT 10000# RJ BORE 160.1MM 2-1/4CR F22M	0	0	
1680	U04	817	FLANGE	5619898	10	10	FLG COMPACT 10000# RJ BORE 201.0MM 2-1/4CR F22M	0	0	
1690	U04	817	FLANGE	5619899	12	12	FLG COMPACT 10000# RJ BORE 239.8MM 2-1/4CR F22M	0	0	
1700	U04	817	FLANGE	5619877	14	14	FLG COMPACT 10000# RJ BORE 263.6MM 2-1/4CR F22M	0	0	
1710	U04	817	FLANGE	5619878	16	16	FLG COMPACT 10000# RJ BORE 302.4MM 2-1/4CR F22M	0	0	
1720	U04	817	FLANGE	5619879	18	18	FLG COMPACT 10000# RJ BORE 341.2MM 2-1/4CR F22M	0	0	
1730	U04	817	FLANGE	5619880	20	20	FLG COMPACT 10000# RJ BORE 379.0MM 2-1/4CR F22M	0	0	
1740	U04	817	FLANGE	5619881	22	22	FLG COMPACT 10000# RJ BORE 417.8MM 2-1/4CR F22M	0	0	
1750			FLANGE	5563412	2	2	FLG WN 10000 API 6A TYPE 6BX 1 13/16" NS (2" NB) RTJ BORE 39.3MM 2-1/4CR F22M	0	0	
1760			FLANGE	5563413	3	3	FLG WN 10000 API 6A TYPE 6BX 2 9/16" NS (3" NB) RTJ BORE 60.9MM 2-1/4CR F22M	0	0	
1770			FLANGE	5563420	4	4	FLG WN 10000 API 6A TYPE 6BX 4 1/16" NS (4" NB) RTJ BORE 80.3MM 2-1/4CR F22M	0	0	
1780			FLANGE	5563421	6	6	FLG WN 10000 API 6A TYPE 6BX 5 1/8" NS (6" NB) RTJ BORE 121.3MM 2-1/4CR F22M	0	0	
1790			FLANGE	5563422	8	8	FLG WN 10000 API 6A TYPE 6BX 7 1/16" NS (8" NB) RTJ BORE 160.1MM 2-1/4CR F22M	0	0	
1800			FLANGE	5563423	10	10	FLG WN 10000 API 6A TYPE 6BX 9" NS (10" NB) RTJ BORE 201.0MM 2-1/4CR F22M	0	0	
1810			FLANGE	5563424	12	12	FLG WN 10000 API 6A TYPE 6BX 11" NS (12" NB) RTJ BORE 239.8MM 2-1/4CR F22M	0	0	
1820			FLANGE	5563425	14	14	FLG WN 10000 API 6A TYPE 6BX 11" (14" NB) NS RTJ BORE 263.6MM 2-1/4CR F22M	0	0	
1830			FLANGE	5563426	16	16	FLG WN 10000 API 6A TYPE 6BX 13 5/8" NS (16" NB) RTJ BORE 302.4MM 2-1/4CR F22M	0	0	
1840			FLANGE	5563427	18	18	FLG WN 10000 API 6A TYPE 6BX 13 5/8" NS (18" NB) RTJ BORE 341.0MM 2-1/4CR F22M	0	0	
1850			FLANGE	5563428	20	20	FLG WN 10000 API 6A TYPE 6BX 16 3/4" NS (20" NB) RTJ BORE 379.0MM 2-1/4CR F22M	0	0	

	TENGINZCHEVROIL TCO TENGINZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - NONE	CORR. ALLOW. 3 MM	SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN CODE ASME B31.3 CHAPTER IX / API 6A DESIGN PRESSURE - 600 BARG FLUID CODES - PIM-SU-512-TCO	MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 10000 RJ/CF TEMPERATURE RANGE:- 40 / 130 °C	12/16/2024	BY OZAC	U04	Page 3 of 5	10000F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1860			FLANGE	5563431	22	22	FLG WN 10000 API 6A TYPE 6BX 16 3/4" NS (22" NB) RTJ BORE 418.0MM 2-1/4CR F22M	0	0	
1870			BLIND	5407422	2	2	FLG BLIND 10000 API 6A TYPE 6BX 1 13/16" NS RTJ 2-1/4CR F22M	0	0	
1880			BLIND	5407423	3	3	FLG BLIND 10000 API 6A TYPE 6BX 2 9/16" NS RTJ 2-1/4CR F22M	0	0	
1890			BLIND	5407430	4	4	FLG BLIND 10000 API 6A TYPE 6BX 4 1/8" NS RTJ 2-1/4CR F22M	0	0	
1900			BLIND	5407431	6	6	FLG BLIND 10000 API 6A TYPE 6BX 5 1/8" NS RTJ 2-1/4CR F22M	0	0	
1910			BLIND	5407432	8	8	FLG BLIND 10000 API 6A TYPE 6BX 7 1/16" NS RTJ 2-1/4CR F22M	0	0	
1920			BLIND	5407433	10	10	FLG BLIND 10000 API 6A TYPE 6BX 9" NS RTJ 2-1/4CR F22M	0	0	
1930			BLIND	5407435	12	14	FLG BLIND 10000 API 6A TYPE 6BX 11" NS RTJ 2-1/4CR F22M	0	0	
1940			BLIND	5407436	16	18	FLG BLIND 10000 API 6A TYPE 6BX 13 5/8" NS RTJ 2-1/4CR F22M	0	0	
1950			BLIND	5407438	20	22	FLG BLIND 10000 API 6A TYPE 6BX 16 3/4" NS RTJ 2-1/4CR F22M	0	0	
1960	U04	817	BLIND	5619755	0.75	1.5	FLG COMPACT BLIND 10000# RJ LTCS A350 LF2 CL.1 LINE CLASS 10000F22 FOR NPS 1 1/2 INCH & BELOW	0	0	
1970	U04	817	BLIND	5619754	2	3	FLG COMPACT BLIND 10000# RJ 2-1/4CR F22M LINE CLASS 10000F22 FOR NPS 2 TO 3 INCH	0	0	
1980	U04	817	BLIND	5619756	4	22	FLG COMPACT BLIND 10000# RJ 2-1/4CR F22M LINE CLASS 10000F22 FOR NPS 4 INCH & ABOVE	0	0	
1990										
1990	U04		FLG ADAPTER	5851295	2	2	FLANGE ADAPTER 10000 API 6A TYPE 6BX 1 13/16 " NS (2" NB) RTJ ALLOY 625 B564 UNS N06625 X 1/2" OD TUBE	0	0	
2000										
2000		FA1					***** ORIFICE FLANGES *****			
2010										
2010			ORIFICE FLG	5407522	2	2	WN ORIF SET 10000 API 6A TYPE 6BX 1 13/16" NS (2" NB) RTJ BORE 39.3MM 2-1/4CR F22M	0	0	
2020			ORIFICE FLG	5407545	14	14	WN ORIF SET 10000 API 6A TYPE 6BX 11" NS (14" NB) RTJ BORE 263.6MM 2-1/4CR F22M	0	0	
2030			ORIFICE FLG	5407546	16	16	WN ORIF SET 10000 API 6A TYPE 6BX 13 5/8" NS (16" NB) RTJ BORE 302.4MM 2-1/4CR F22M	0	0	
2040			ORIFICE FLG	5407547	18	18	WN ORIF SET 10000 API 6A TYPE 6BX 13 5/8" NS (18" NB) RTJ BORE 341.0MM 2-1/4CR F22M	0	0	
2050			ORIFICE FLG	5407711	22	22	WN ORIF SET 10000 API 6A TYPE 6BX 16 3/4" NS (22" NB) RTJ BORE 418.0MM 2-1/4CR F22M	0	0	
2060										
2060							*****PIPE NIPPLES*****			
2070										
2070		812		0			NIPPLE PBE CUT FROM PIPE			
2080										
2080		FA1					***** GASKETS *****			
2090										
2090			GASKET	5674471	2	2	GASKET 10000 API 6A BX 151 1 13/16" NS 316 SS RING	0	0	
2100			GASKET	5674475	3	3	GASKET 10000 API 6A BX 153 2 9/16" NS 316SS RING	0	0	
2110			GASKET	5674498	4	4	GASKET 10000 API 6A BX 155 4 1/16" NS 316 SS RING	0	0	
2120			GASKET	5674497	6	6	GASKET 10000 API 6A BX 169 5 1/8" NS 316SS RING	0	0	
2130			GASKET	5674486	8	8	GASKET 10000 API 6A BX 156 7 1/16" NS 316SS RING	0	0	
2140			GASKET	5674487	10	10	GASKET 10000 API 6A BX 157 9" NS 316SS RING	0	0	
2150			GASKET	5674483	12	14	GASKET 10000 API 6A BX 158 11" NS 316SS RING	0	0	
2160	U04		GASKET	5674484	16	18	GASKET 10000 API 6A BX 159 13 5/8" NS 316SS RING	0	0	
2170	U04		GASKET	5674488	20	22	GASKET 10000 API 6A BX 162 16 3/4" NS 316SS RING	0	0	
2180	U04	817	GASKET	5673554	0.5	3	GASKET 10000# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 10000F22 FOR NPS 3 INCH & BELOW	0	0	
2190	U04	817	GASKET	5673554	0.75	3	GASKET 10000# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 10000F22 FOR NPS 3 INCH & BELOW	0	0	
2200	U04	817	GASKET	5673550	4	22	GASKET 10000# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 10000F22 FOR NPS 4 INCH & ABOVE	0	0	
2210										
2210							***** BOLTING *****			
2220										
2220	U04	BA3	BOLTING	5675770	0.5	18	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
2230		BA5	BOLTING	5675778	6	99	STUD-BOLT C/W 2 COMPACT NUTS API-6A718	0	0	
2240		BA3	BOLTING	5675776	20	99	STUD-BOLT WITH 2 NUTS API-6A718	0	0	
2250										
2250		SB6	BOLTING	5675675	8	22	JACK SCREW HEX HEAD A320 GR L7M	0	0	
2260										
2260		FA1					***** BLINDS & SPACERS *****			
2270										
2270			SPEC BLIND	5407622	2	2	SPECTACLE BLIND 10000 API 6A 1 13/16" NS FOR 2" PIPE RTJ FEMALE 2-1/4CR F22M	0	0	
2280			SPEC BLIND	5407623	3	3	SPECTACLE BLIND 10000 API 6A 2 9/16" NS FOR 3" PIPE RTJ FEMALE 2-1/4CR F22M	0	0	
2290			SPEC BLIND	5407624	4	4	SPECTACLE BLIND 10000 API 6A 4 1/16" NS FOR 4" PIPE RTJ FEMALE 2-1/4CR F22M	0	0	
2300										
2300			PADDLE	5407742	8	8	PADDLE BLANK 10000 API 6A 7 1/16" NS FOR 8" PIPE RTJ FEMALE 2-1/4CR F22M	0	0	
2310			PADDLE	5407842	8	8	PADDLE SPACER 10000 API 6A 7 1/16" NS FOR 8" PIPE RTJ FEMALE 2-1/4CR F22M	0	0	
2320			PADDLE	5407745	10	10	PADDLE BLANK 10000 API 6A 9" NS FOR 10" PIPE RTJ FEMALE 2-1/4CR F22M	0	0	
2330			PADDLE	5407845	10	10	PADDLE SPACER 10000 API 6A 9" NS FOR 10" PIPE RTJ FEMALE 2-1/4CR F22M	0	0	
2340			PADDLE	5407747	14	14	PADDLE BLANK 10000 API 6A 11" NS FOR 14" PIPE RTJ FEMALE 2-1/4CR F22M	0	0	
2350			PADDLE	5407847	14	14	PADDLE SPACER 10000 API 6A 11" NS FOR 14" PIPE RTJ FEMALE 2-1/4CR F22M	0	0	
2360			PADDLE	5407746	16	16	PADDLE BLANK 10000 API 6A 13 5/8" NS FOR 16" PIPE RTJ FEMALE 2-1/4CR F22M	0	0	
2370			PADDLE	5407846	16	16	PADDLE SPACER 10000 API 6A 13 5/8" NS FOR 16" PIPE RTJ FEMALE 2-1/4CR F22M	0	0	
2380			PADDLE	5407744	18	18	PADDLE BLANK 10000 API 6A 13 5/8" NS FOR 18" PIPE RTJ FEMALE 2-1/4CR F22M	0	0	
2390			PADDLE	5407844	18	18	PADDLE SPACER 10000 API 6A 13 5/8" NS FOR 18" PIPE RTJ FEMALE 2-1/4CR F22M	0	0	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - NONE	CORR. ALLOW. 3 MM	SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN CODE ASME B31.3 CHAPTER IX / API 6A DESIGN PRESSURE - 600 BARG FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 10000 RJ/CF TEMPERATURE RANGE: -40 / 130 °C	12/16/2024	BY OZAC	U04	Page 4 of 5	10000F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
2400			PADDLE	5407784	22	22	PADDLE BLANK 10000 API 6A 16 3/4" NS FOR 22" PIPE RTJ FEMALE 2-1/4CR F22M	0	0	
2410			PADDLE	5407884	22	22	PADDLE SPACER 10000 API 6A 16 3/4" NS FOR 22" PIPE RTJ FEMALE 2-1/4CR F22M	0	0	
2420										
2420	U04	817	SPEC BLIND	5749571	2	3	SPECTACLE BLIND COMPACT 10000# RJ FEMALE 2-1/4CR F22M PIPE CLASS 10000F22 FOR NPS 3 INCH & BELOW	0	0	
2430	U04	817	SPEC BLIND	5749576	4	4	SPECTACLE BLIND COMPACT 10000# RJ FEMALE 2-1/4CR F22M PIPE CLASS 10000F22 FOR NPS 4 INCH	0	0	
2440										
2440	U04	817	PADDLE	5750294	2	3	PADDLE BLANK COMPACT 10000# RJ 2-1/4CR F22M LINE CLASS 10000F22 FOR NPS 3 INCH & BELOW	0	0	
2450	U04	817	PADDLE	5750295	2	3	PADDLE SPACER COMPACT 10000# RJ 2-1/4CR F22M LINE CLASS 10000F22 FOR NPS 3 INCH & BELOW	0	0	
2460	U04	817	PADDLE	5750276	8	22	PADDLE BLANK COMPACT 10000# RJ 2-1/4CR F22M LINE CLASS 10000F22	0	0	
2470	U04	817	PADDLE	5750276	4	22	PADDLE BLANK COMPACT 10000# RJ 2-1/4CR F22M PIPE CLASS 10000F22	0	0	
2480	U04	817	PADDLE	5750288	4	22	PADDLE SPACER COMPACT 10000# RJ 2-1/4CR F22M PIPE CLASS 10000F22	0	0	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - NONE	CORR. ALLOW. 3 MM	SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN CODE ASME B31.3 CHAPTER IX / API 6A DESIGN PRESSURE - 600 BARG FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 10000 RJ/CF TEMPERATURE RANGE: -40 / 130 °C	12/16/2024	BY OZAC	U04	Page 5 of 5	10000F22